

Objective

Build and deploy a **full-stack MERN mini-CRM application** with secure authentication, CRUD operations, protected routes, and cloud deployment.

Authentication Module

Build a complete authentication flow:

Requirements

- Sign Up
- Sign In
- JWT-based authentication
- Password hashing (bcrypt)
- Protect backend routes using middleware
- Protect frontend routes (redirect if not logged in)
- Token stored securely
- Token refresh mechanism

CRM Module

After login, the user should access a **Contacts Dashboard**.

Contact Fields

- Name
- Email
- Phone
- Company
- Status (Lead / Prospect / Customer)
- Notes
- CreatedAt / UpdatedAt

Features

- Add a contact
- Edit a contact
- Delete a contact
- View contacts list
- Search (by name or email) OR Filter by status (choose at least one)

Frontend Requirements (React)

- Responsive layout
- Authentication pages (Login, Signup)
- Contacts dashboard page

- Form validation
- Protected routes
- Pagination (**required**) — 10 items per page
- Activity logs (add/edit/delete events)
- Unit Test cases
- CSV export for contacts (optional)
- Role-based access (Admin/User) (optional)

Backend Requirements (Node.js + Express + MongoDB)

- Proper folder structure (controllers, routes, models, middleware)
- REST API endpoints for auth + contacts
- Input validation
- Error handling with correct HTTP codes
- Login rate limit (3 requests per 10 min)
- Mongoose pagination
- Better schema validation
- Unit tests
- Dockerized backend (optional)

Deployment

Deploy both frontend and backend using any free cloud platform.

Deliverables

- Live frontend URL
- Live backend API URL
- Postman Collection OR API documentation
- GitHub repos (frontend + backend)
- README containing:
 - Setup instructions
 - Architecture explanation

Final Submission Checklist

- ✓ Live deployed app
- ✓ Live deployed backend API
- ✓ GitHub repositories
- ✓ README with documentation
- ✓ Postman collection or API docs